

Solitons in a Nonlinear Inhomogeneous Transmission Line

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Nondestructive propagation of solitons is described in a nonlinear, electrical transmission line having loss and inhomogeneity. According to its nonlinear property expressed in terms of the $[P+1]$ -th power, the transmission line demands a certain balance between the magnitude of loss or activity and inhomogeneity to preserve a unique soliton form of $\text{sech}^{2/P}$ with exponentially varying amplitude. The adiabatic amplification effect for waves propagating in the line is considered by estimating their conserved quantities. The theoretical anticipation is confirmed by several numerical calculations. It is also shown that, for $P \geq 4$, it seems that solitons never travel stably in the real nonlinear transmission line.

§1. Introduction

In this decade, various transmission line models for solitons and related phenomena have been discussed in many articles.^{1,2)} The recent advance is, in particular, some discovery of the transmission line models which can be shown to have the exact N -interacting soliton solution. For the first time Hirota and Suzuki³⁾ showed that a certain nonlinear network is an electrical model analog to the well known Toda's exponential lattice⁴⁾ in dynamics. Later, other different networks which could be treated analytically were investigated by Hirota and Satsuma,⁵⁾ and others.⁶⁻⁸⁾ However, as a solvable model, each section of such ladder networks has to contain both nonlinear inductor and capacitor which indicate peculiar characteristics of nonlinear mathematical functions.

In parallel with above theoretical investigations another conventional approach has been performed to get a good description of nonlinear phenomena in the realistic nonlinear transmission lines. Noguchi⁹⁾ applied the perturbation method¹⁰⁾ to analyze an electrical ladder network of the low-pass filter, and found to explain the comparatively low frequency nonlinear waves such as video solitons in the

line. According to this perturbation method, the low frequency waves in the lossless transmission line with $[P+1]$ -th power nonlinearity can be well approximated in the P -th nonlinear K-dV equation.^{9,11)} In other words, the low frequency nonlinear waves are described by the fundamentals of both linearized dispersion relation and nonlinear property of the transmission line.

The main aim of this article is to extend this method to a case of lossy and inhomogeneous nonlinear transmission lines, and to discuss some interesting consequences. We can find several studies on this problem, *e.g.*, a lossless but linearly inhomogeneous case,¹²⁾ a randomly inhomogeneous case,¹³⁾ and a nonlinear transmission line with small dissipations.¹⁴⁾ In these examples, however, the describing equations could not have any analytical solutions as puls-like waves, and so experimental or numerical treatments were used in their estimations.

We restrict to an exactly solvable case of a lossy and inhomogeneous transmission line with $[P+1]$ -th power nonlinearity. In the next section, we introduce a model of the nonlinear transmission line, and derive an approximate equation which describes the comparatively low frequency nonlinear waves. We also discuss on the particular case of the approximate equation called the P -th nonlinear K-dV equation with loss and inhomogeneity. Section 3 is the most interesting one in which

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we consider about the properties of soliton solutions, and use numerical calculations to compare with the corresponding theoretical results. Section 4 is devoted to more careful discussions on the mechanism of the amplification or damping of video solitons. The last section is a conclusion.

§2. Derivation of a Modified Type of the K-dV Equation

We consider an infinitely long transmission line (or ladder networks) composed of lumped sections; a typical section is shown in Fig. 1. If $\Phi_n(i_n)$ is the nonlinear flux per unit length Δ at stage n , the flux of the section is $\Delta\Phi_n(i_n)$ webers. Similarly the resistance of the section is ΔR_n ohms, the nonlinear charge is $\Delta Q_n(v_n)$ coulombs, the leakage conductance is ΔG_n mohs.

The fundamental equations derived from Fig. 1 are

$$i_n - i_{n+1} = \Delta G_n v_n + \Delta \frac{\partial Q_n}{\partial t}, \quad (1)$$

$$v_{n-1} - v_n = \Delta R_n i_n + \Delta \frac{\partial \Phi_n}{\partial t}. \quad (2)$$

We express each parameters of the line as follows:

$$\Phi_n = L_0 [1 - \Gamma_L(n, t) - \alpha_L i_n^P / (P+1)] i_n, \quad (3)$$

$$Q_n = C_0 [1 - \Gamma_C(n, t) - \alpha_C v_n^P / (P+1)] v_n, \quad (4)$$

$$R_n = R_0 [1 - \Gamma_R(n, t)], \quad (5)$$

$$G_n = G_0 [1 - \Gamma_G(n, t)], \quad (6)$$

$$f_{n\pm 1}(t) = \left[f(x, t) \pm \Delta \frac{\partial f}{\partial x} + \frac{\Delta^2}{2} \frac{\partial^2 f}{\partial x^2} \pm \frac{\Delta^3}{6} \frac{\partial^3 f}{\partial x^3} + \frac{\Delta^4}{24} \frac{\partial^4 f}{\partial x^4} \pm \dots \right]_{x=n\Delta}, \quad (9)$$

rewriting i_n , v_n , i_{n+1} and v_{n-1} in eqs. (1) and (2), we obtain a set of nonlinear partial differential equation for $v(x, t)$ and $i(x, t)$. From the Ohm's law, if we can assume

$$v \simeq \left[\frac{\partial \Phi}{\partial i} / \frac{\partial Q}{\partial v} \right]^{1/2} i \simeq \left(\frac{L_0}{C_0} \right)^{1/2} i = Zi, \quad (10)$$

and introduce $u(x, t)$ as the right progressing waves;

$$u = Z^{1/2} i = Z^{-1/2} v, \quad (11)$$

the next two equations are derived as,

$$\frac{\partial u}{\partial x} + \frac{\Delta}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta^2}{6} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta^3}{24} \frac{\partial^4 u}{\partial x^4} + \frac{\Delta^4}{120} \frac{\partial^5 u}{\partial x^5} + \dots + RZu + Z \frac{\partial Q(Z^{1/2}u)}{\partial t} = 0, \quad (12)$$

$$\frac{\partial u}{\partial x} - \frac{\Delta}{2} \frac{\partial^2 u}{\partial x^2} + \frac{\Delta^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{\Delta^3}{24} \frac{\partial^4 u}{\partial x^4} + \frac{\Delta^4}{120} \frac{\partial^5 u}{\partial x^5} - \dots + GZ^{-1}u + Z^{-1} \frac{\partial \Phi(Z^{-1/2}u)}{\partial t} = 0. \quad (13)$$

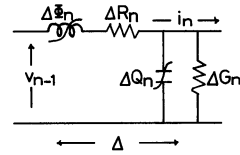


Fig. 1. A nonlinear transmission line, (a unit section).

$$\Gamma_L / \Gamma_{L_0} = \Gamma_C / \Gamma_{C_0} = \Gamma_R / \Gamma_{R_0} = \Gamma_G / \Gamma_{G_0} = \Gamma(n, t). \quad (7)$$

Where $\Gamma(n, t)$ describes an inhomogeneity of the line. We assume that $\Gamma(n, t)$ is a slowly varying function with respect both n and t and so,

$$\begin{aligned} \Gamma(n, t) &= \Gamma(n\Delta - (L_0 C_0)^{-1/2} t) = \Gamma(\xi) \\ &= \Gamma(\xi_0 + (\xi - \xi_0)) \\ &\simeq \Gamma(\xi_0) + (\xi - \xi_0) \left. \frac{\partial \Gamma}{\partial \xi} \right|_{\xi=\xi_0}, \end{aligned} \quad (8)$$

and the magnitudes of dissipations (R_0/G_0) are assumed to be both small. The nonlinear inhomogeneous transmission line shown here slightly differs from the previous line¹⁵⁾ which has the frequency dispersion even in the limit of distributed network ($\Delta \rightarrow 0$). In the examination of the present transmission line we must remember that the line is constructed from lumped parameters ($\Delta = 1$). So we consider the continuum approximation of eqs. (1) and (2). This is a reasonable approximation for the long wavelength (or low frequency) waves comparing with the length of a section of the line. By using

Adding eqs. (12) and (13), and neglecting small terms,

$$2\frac{\partial u}{\partial x} + \frac{A^2}{3}\frac{\partial^3 u}{\partial x^3} + (GZ + RZ^{-1})u + Z\frac{\partial Q}{\partial t} + Z^{-1}\frac{\partial \Phi}{\partial t} = 0. \quad (14)$$

In eq. (14), we introduce a coordinate system moving to the right with the velocity of an infinitesimally small amplitude wave along the line,

$$\xi = x - (L_0 C_0)^{-1/2}t, \quad \tau = (L_0 C_0)^{-1/2}t, \quad (15)$$

and using the simple relation; $\partial/\partial\tau \simeq -\partial/\partial\xi$, we get the following modified type of the P-th nonlinear K-dV equation having loss and nonuniformity terms,

$$\frac{\partial u}{\partial \tau} + \alpha u^P \frac{\partial u}{\partial \xi} + \beta \frac{\partial^3 u}{\partial \xi^3} + \gamma \frac{\partial u}{\partial \xi} + \delta u = 0, \quad (16)$$

where

$$\alpha = \alpha_C Z^{P/2} + \alpha_L Z^{-P/2}, \quad \beta = \frac{A^2}{6} = \frac{1}{6},$$

$$\gamma = \Gamma_C + \Gamma_L = (\gamma_{C_0} + \gamma_{L_0})\Gamma(\xi) = \gamma_0\Gamma(\xi), \quad \delta = \gamma_0 + (GZ + RZ^{-1})/2. \quad (17)$$

The above eq. (16) with eqs. (17) seems to describe well the comparatively long wavelength waves in the line. Hereafter, we examine a simpler case of which the previous assumption (10) is completely valid;

$$\gamma_{C_0} = \gamma_{L_0} = \gamma_0/2, \quad \alpha_L Z^{-P/2} = \alpha_C Z^{P/2} = \alpha_0/2, \quad \text{and} \quad G = R = d_0/(Z + Z^{-1}) = \text{const.} \quad (18)$$

The coefficients of eq. (16), eqs. (17) are then,

$$\alpha = \alpha_0, \quad \beta = 1/6, \quad \gamma = \gamma_0(\xi - \xi_0), \quad \text{and} \quad \delta = \gamma_0 + d_0/2. \quad (19)$$

Besides the above critical case, even in the below cases,

$$|\alpha_C Z^{P/2} - \alpha_L Z^{-P/2}|, \quad |\gamma_{C_0} - \gamma_{L_0}|, \quad |\gamma_{G_0}Z + \gamma_{R_0}Z^{-1}| = O(\varepsilon), \quad (20)$$

the assumption (10) and so eq. (16) with eqs. (17) may still hold sufficiently well.¹⁵⁾

Here, we introduce the following dependent variable transformations;

$$U(\zeta, \theta) = \exp(2\gamma_0\tau/P)u(\xi, \tau),$$

$$\theta = \exp(-3\gamma_0\tau)/(-3\gamma_0),$$

$$\zeta = \exp(-\gamma_0\tau)(\xi - \xi_0), \quad (21)$$

and rewrite eq. (16) to the P-th nonlinear K-dV equation with an additional term,

$$\frac{\partial U}{\partial \theta} + \alpha U^P \frac{\partial U}{\partial \zeta} + \beta \frac{\partial^3 U}{\partial \zeta^3} + \sigma \theta^{-1} U = 0, \quad (22)$$

where $\sigma = 2/3P - \delta/3\gamma_0$. Figure 2 indicates the conditions in which the last term of eq. (22) disappears ($\sigma=0$) for the various nonlinear property P of the line. In such cases, eq. (22) reduces to the usual P-th nonlinear K-dV equation which has the single soliton solution to each natural number P. We discuss in detail on the property of these soliton solutions in the next section.

In the final of this section, we denote some

other solvable case without the condition $\sigma=0$. If $\sigma=1$ ($\sigma=2$), eq. (22) reduces to the P-th nonlinear spherical (cylindrical) K-dV equation. For P=1, such spherical (cylindrical) soliton like pulses have been already found as ion-acoustic plasma waves.^{16,17)} It is interesting that the behavior of three dimensional

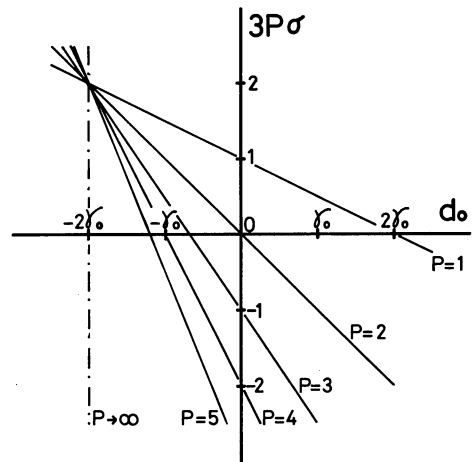


Fig. 2. Schematic description of conditions $\sigma=0$.

nonlinear waves can be simulated in the one-dimensional nonlinear transmission line with loss and nonuniformity.

In general, by introducing the transformation;¹⁸⁾

$$U' = U \exp \left(\sigma \int \theta^{-1} d\theta \right), \quad (23)$$

and substituting to eq. (22), then

$$\frac{\partial U'}{\partial \theta} + \alpha'(\theta) U'^P \frac{\partial U'}{\partial \xi} + \beta \frac{\partial^3 U'}{\partial \xi^3} = 0, \quad (24)$$

where

$$\alpha'(\theta) = \alpha \exp \left(-P\alpha \int \theta^{-1} d\theta \right).$$

The asymptotic solutions of the above variable-coefficient K-dV equation (24) for $P=1$ and 2 have been discussed in detail,^{19,20)} so we can obtain a little more information on the solutions of eq. (24), and so of eq. (16).

§3. Properties of Solitons

In this section, we show the single soliton solution of the P -th nonlinear K-dV equation with loss and nonuniformity terms (16). The single soliton solution is derived from that of the usual K-dV equation (22), through the explicit transformations (21), under the condition of

$$\sigma = 2/3P - \delta/3\gamma_0 = 0, \quad (25)$$

as

$$u(\xi, \tau) = u_s(\tau) \operatorname{sech}^{2/P} \times \left[W_s^{-1}(\tau) \left\{ \xi - \xi_0(1 - e^{\gamma_0 \tau}) + \frac{V_s(\tau)}{3\gamma_0} (1 - e^{3\gamma_0 \tau}) \right\} \right], \quad (26)$$

where

$$\begin{aligned} u_s(\tau) &= u_0 \exp(-2\gamma_0 \tau/P), \\ V_s(\tau) &= 2\alpha u_s(\tau)^P/(P+1)(P+2), \\ W_s(\tau) &= [P^2 V_s(\tau)/4\beta]^{-1/2}. \end{aligned}$$

The amplitude dependent velocity of the single soliton (26) is

$$\frac{d\xi}{d\tau} = -\frac{\partial \phi}{\partial \tau} \bigg/ \frac{\partial \phi}{\partial \xi} \gamma_0 (\xi - \xi_0) + V_s(\tau) \exp(-2\gamma_0 \tau), \quad (27)$$

where $\phi(\xi, \tau)$ is the phase of $\operatorname{sech}^{2/P}[\phi]$ in eq. (26). This velocity becomes a constant value $V_s(\tau)|_{\gamma_0=0}$ to each natural number P . For any P , if $\gamma_0 > 0$ ($\gamma_0 < 0$), then the single soliton comes to broaden (compress) its own width during the propagation, since the front (tail) part of the body moves faster than tail (front). In Fig. 3, 4, and 5, we show typical results of numerical simulation of eq. (16) for $P=1$, 2, and 3, respectively. We used the single soliton-form (26) as the initial wave profile, and other experimental conditions were denoted in the corresponding figure captions. As is already shown in eq. (25), to preserve a unique soliton form without emitting any oscillatory waves, the following relations must be satisfied between the magnitudes of inhomogeneity and loss or activity;

$$\begin{aligned} \gamma_0 &= (GZ + RZ^{-1})/2 \quad \text{for } P=1, \\ GZ + RZ^{-1} &= 0 \quad \text{and any } \gamma_0 \quad \text{for } P=2, \\ \gamma_0 &= -(GZ + RZ^{-1})/(2-4/P) \quad \text{for } P \geq 3. \end{aligned}$$

It is evaluated that, for the cases of $P \geq 2$, the amplitudes of solitons are grown ($\gamma_0 < 0$) by the inhomogeneity even in the transmission

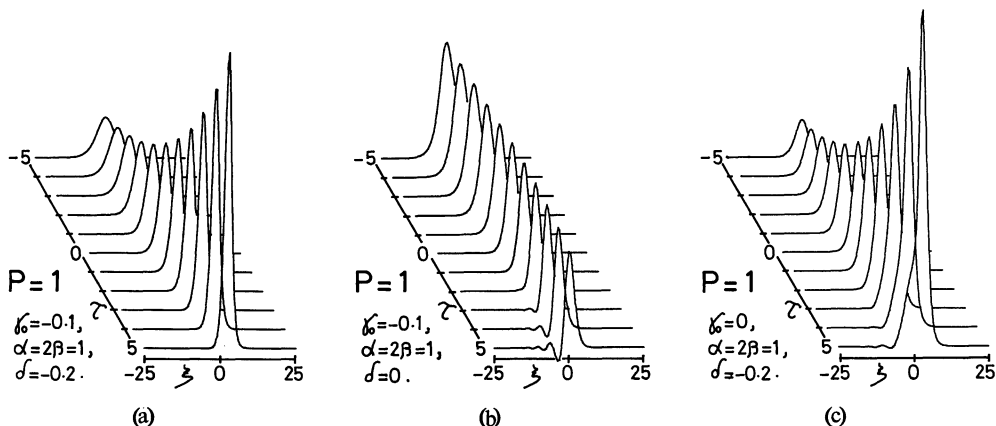


Fig. 3. Amplification and damping of one-soliton for $P=1$; (a) $\sigma=0$, ($\gamma_0=-0.1$, $\delta=-0.2$), (b) $\sigma \neq 0$, ($\gamma_0=-0.1$, $\delta=0$), (c) $\sigma \neq 0$, ($\gamma_0=0$, $\delta=-0.2$).

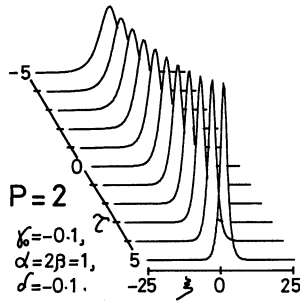


Fig. 4. Amplification and damping of one-soliton for $P=2$.

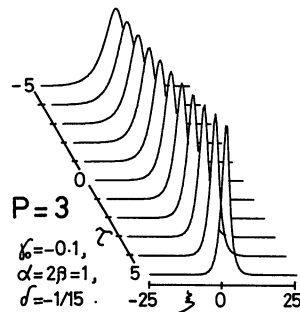


Fig. 5. Amplification and damping of one-soliton for $P=3$.

line having real dissipations.

We examine such adiabatic amplification effect by a gradual inhomogeneity of parameters. As will be shown in the next section, the P -th nonlinear K-dV equation possesses at least three polynomial conserved quantities. The instantaneous energy of waves propagating in the line is proportional to u^2 , because of the assumption (11). By using the result (eq. (32)) of the next section, the total energy of the single soliton (26) is derived in the moving frame as,

$$E_P(\tau) = \int_{-\infty}^{\infty} \left(\frac{u^2}{2} \right) d\xi = \int \frac{1}{2} u_s^2 \operatorname{sech}^{4/P}[\phi] d\xi \\ = \frac{A_P}{2} u_s^2 W_s, \quad (28)$$

where

$$A_P = \int_{-\infty}^{\infty} \operatorname{sech}^{4/P} \xi' d\xi', \quad A_1 = 4/3, \\ A_2 = 2, \quad A_3 \approx 2.587, \quad \text{and} \quad A_4 = \pi.$$

The amplitude u_0 and time dependences of the above total energy E_P are calculated as,

$$E_P(\tau) \propto u_0^{2-P} \exp[(1-4/P)\gamma_0\tau]. \quad (29)$$

We note that, for $P > 4$, eq. (29) shows the

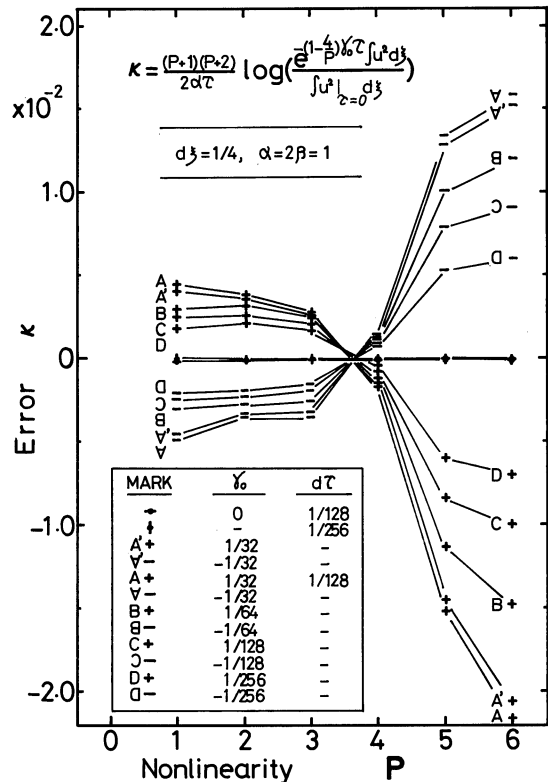


Fig. 6. An estimation of the errors due to instability of one-soliton.

total energy of soliton comes to depend negatively on its amplitude, and the time dependence of the energy also changes. The total energy of the single soliton comes to decrease with the passage of time even for $\gamma_0 < 0$ against the growth of the amplitude of soliton. For $P=4$, the energy of the soliton becomes a constant value with respect to both amplitude and time. Consequently, it seems that such anomalous dependences of the total energy for the occasions of $P \geq 4$, (i.e., $\partial E_P / \partial u_0 \leq 0$) never permit the corresponding single soliton to propagate stably in the nonlinear transmission line. We conclude this section by showing a brief confirmation on the above conjecture in Fig. 6. The figure indicates that the obvious difference between two groups, $P \leq 3$ and $P > 4$ in the range of stable propagation of the single solitons.

§4. Adiabatic Amplification and Conserved Quantities

It is well known that the usual K-dV equation and the modified K-dV equation (in this paper,

they correspond to the cases $P=1$ and 2 of eq. (22) with $\sigma=0$, respectively) have the N -soliton solutions^{21,22)} which denote the time and space evolution of N -multiple interacting solitons. So, we can easily obtain the N -soliton solutions of eq. (16) by using already known N -soliton for $P=1$ and 2 through our transformations (21).

The N -soliton solutions $U(\zeta, \theta)$ for $P=1$ and 2 are described as follows:²¹⁻²⁴⁾

$$U^P = \frac{2\beta(P+1)(P+2)}{\alpha P^2} \frac{\partial^2}{\partial \zeta^2} [\log w], \quad w = \det |H|,$$

$$H_{ij} = \delta_{ij} + \left[\frac{2(k_i k_j)^{1/2}}{k_i + k_j} \right] \exp \left(\frac{\zeta_i + \zeta_j}{2} \right),$$

$$\zeta_m = k_m \zeta - \beta k_m^3 \theta - \theta_m, \quad m = 1, 2, 3, \dots, N, \quad (30)$$

where δ_{ij} is the Kronecker's delta and θ_m constants. It should be noted again that the description (30) is valid only for $P=1$ and 2 . As an example, in Fig. 7, we show the configuration of 3-interacting solitons for $P=2$. Although, we can see the effect of amplification or damping in Fig. 7, from the condition (25), in this case the nonlinear transmission line may not have any activities nor losses. Such adiabatic (or generally, parametric) amplification or damping is understood by estimating the following conserved quantities

$$\begin{aligned} I_3 &= \int_{-\infty}^{\infty} \frac{1}{P+1} \left[U^{P+2} - \frac{(P+1)(P+2)\beta}{2\alpha} \left(\frac{\partial U}{\partial \zeta} \right)^2 \right] d\zeta \\ &= e^{(4/P+1)\gamma_0 \tau} \int_{-\infty}^{\infty} \frac{1}{P+1} \left[u^{P+2} - \frac{(P+1)(P+2)\beta}{2\alpha} \left(\frac{\partial u}{\partial \xi} \right)^2 \right] d\xi. \end{aligned} \quad (33)$$

These quantities are considered to hold as the conservation of momentum, energy, and another higher order constant of motion for each P . Only for $P=1$ and 2 , further higher order (infinite) conserved quantities have been derived.^{23,24)}

We show the next higher order constant of motion as an example of the use of the transformations (21),

$$\begin{aligned} I_4 &= \int_{-\infty}^{\infty} \frac{1}{2(P+1)} \left[U^{2(P+1)} - \frac{2(P+1)(2P+1)\beta}{\alpha} U^P \left(\frac{\partial U}{\partial \zeta} \right)^2 + \frac{6(P+1)(2P+1)\beta^2}{5\alpha^2} \left(\frac{\partial^2 U}{\partial \zeta^2} \right)^2 \right] d\zeta \\ &= e^{(4/P+3)\gamma_0 \tau} \int_{-\infty}^{\infty} \frac{1}{2(P+1)} \left[u^{2(P+1)} - \frac{2(P+1)(2P+1)\beta}{\alpha} u^P \left(\frac{\partial u}{\partial \xi} \right)^2 + \frac{6(P+1)(2P+1)\beta^2}{5\alpha^2} \left(\frac{\partial^2 u}{\partial \xi^2} \right)^2 \right] d\xi, \end{aligned} \quad (34)$$

for $P=1$ and 2 . Since the interaction of 3-solitons for $P=2$ is known to be described by the fifth conserved quantity itself, the existence of the corresponding integral, which depends only on time as $\exp(-7\gamma_0 \tau)$, is indirectly indicated by the fact that famous particle-like

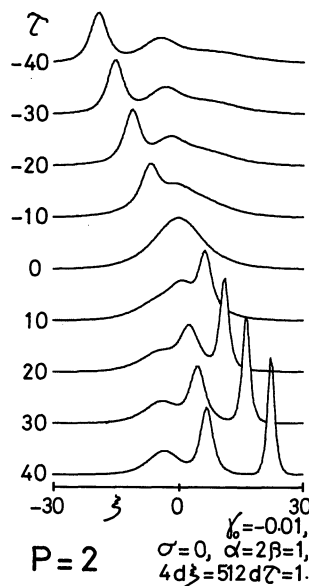


Fig. 7. Three interacting solitons for $P=2$, ($\sigma=0$).

(or constants) of motion. To any natural number P , the first three conserved quantities are obtained as:

$$I_1 = \int_{-\infty}^{\infty} U d\zeta = e^{(2/P-1)\gamma_0 \tau} \int_{-\infty}^{\infty} u d\xi, \quad (31)$$

$$I_2 = \int_{-\infty}^{\infty} \frac{U^2}{2} d\zeta = e^{(4/P-1)\gamma_0 \tau} \int_{-\infty}^{\infty} \frac{u^2}{2} d\xi, \quad (32)$$

interaction of solitons can still hold as shown in Fig. 7.

§5. Concluding Remarks

In this paper, it is shown that the P -th power nonlinear transmission lines with a certain

loss and inhomogeneity can still support the single solitons to each natural number P , and especially to $P=1$ and 2, even the exact N -multi interacting solitons. We have obtained a generalization of the result of the previous report¹⁵⁾ in which a special case as eq. (20) in this article had been treated. The theoretical conditions of balancing between the magnitudes of loss and inhomogeneity to each nonlinearity P show that the nondestructive amplification or damping of solitons without radiations may seem to be done in the P -th power nonlinear transmission lines with loss or activity and inhomogeneity.

Although the stability analysis on the propagating single solitons shown in the last half of section 3 is considered not to be completed, a numerical result briefly indicates that, in the transmission line with P -th ($P \geq 4$) nonlinearity, we can not only amplify nor reduce the amplitude of soliton nondestructively but even see the stable propagation of the single soliton. The similar method for the stability analysis of the single soliton have been already reported by several authors,^{25,26)} and we can see their results for the K-dV equation and a different nonlinear evolution equation bears very well resemblance to ours.

It is not difficult to extend our result for the nonlinearity $P=1$ to a shallow water waves,²⁷⁾ a nonlinear lattice waves,²⁸⁾ and a few types of nonlinear plasma waves.²⁹⁾ The Alfvén waves³⁰⁾ in a nonuniform plasma may be also described by our modified type of the K-dV equation (16) with the nonlinearity $P=2$.

As well as the amplification or damping of solitons, it is also interesting to consider nonlinear periodical waves such as the cnoidal waves in inhomogeneous media. The well known recurrence phenomenon brought up by Fermi, Pasta, and Ulam³¹⁾ has been also observed in nonlinear transmission lines having quadratic ($P=1$)^{32,33)} and cubic ($P=2$)³²⁾ nonlinearity. In conjunction with the problem of thermalization, the recurrence like phenomenon in the nonlinear inhomogeneous media is considered to be worth studying.

As a final remark, to realize the nondestructive amplification or damping of amplitude of solitons in the transmission line having the nonlinearity $P=2$, we can use a coupling between high and low frequency waves pro-

pagating together in the line. The method and result will be appeared in the next paper.

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